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Does generative AI copy? Rethinking the right to copy under copyright law

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ABSTRACT

Copyright-regulated reproduction should encompass both technological and economic elements. The technological element, which means the generation of a reproduction, determines whether a certain act constitutes reproduction. The economic element, which means the potential for public distribution and undermining copyright owners' incentives by the generated reproduction, ascertains whether such an act of reproduction falls within copyright law's jurisdiction. We not only delineate the boundary of copyright-regulated reproduction but also elucidate the underlying rationale for the spectrum of reproduction regulated by presumption, reproduction regulated by exemption, and non-regulated reproduction. Accordingly, we analyze the regulatability of GenAI's reproduction of works throughout its stages. During the data acquisition and preprocessing stages, GenAI engages in reproduction with the technological element but lacks the economic element. During the training and generation stages, GenAI normally does not generate output similar to training data, thus initially lacking the technological element; in exceptional cases where GenAI generates output similar to prior works and possesses the technological element, copyright-regulated reproduction only occurs in the generation stage where the output has the potential for public distribution. Furthermore, we address the possible criticism that GenAI's unregulated reproduction would lead to an inequitable scenario by free riding on preexisting works.

1. Introduction

The development of generative AI (GenAI) has brought about disruptive changes to society and posed significant challenges for the legal system,[1] particularly in the realm of copyright law.[2] GenAI is trained on an enormous amount of data, which may be protected by copyright. This has raised considerable debate regarding how to regulate GenAI's use of works through copyright law.[3] Currently, three primary perspectives exist on this issue. First, GenAI can use works without authorization for free since its use can generally be exempted by fair use[4] or text and data mining (TDM) exceptions.[5] Second, GenAI constitutes infringement unless it acquires authorization from copyright

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[1] Bart van der Sloot, *Regulating the Synthetic Society: Generative AI, Legal Questions, and Societal Challenges* (Hart Publishing 2024) 91-92.
[2] Mark A. Lemley, 'How Generative AI Will Turn Copyright on its Head' (2024) 25(2) Columbia Science & Technology Law Review 190.
[3] Pamela Samuelson, 'Generative AI Meets Copyright' (2023) 381 Science 158.
[4] Mark A. Lemley & Bryan Casey, 'Fair Learning' (2021) 99 Texas Law Review 743, 776; Matthew Sag, 'Fairness and Fair Use in Generative AI' (2024) 92(5) Fordham Law Review 1887, 1921; Samuelson (n 3) 161.
[5] Kalpana Tyagi, 'Copyright, Text & Data Mining and the Innovation Dimension of Generative AI' (2024) 19(7) Journal of Intellectual Property Law & Practice 557, 562 ('Central to the entire value chain of Generative AI is the process of machine reading of large volume of data to discover patterns, generate new knowledge and extract insights. This process is referred to as TDM.'); Artha Dermawan, 'Text and Data Mining Exceptions in the Development of Generative AI Models: What the EU Member States could Learn from the Japanese "Nonenjoyment" Purposes?' (2024) 27(1) The Journal of World Intellectual Property 44, 59 (arguing that 'the full implementation of the TDM exceptions would be critical to European innovation and research, particularly, in the development of AI-driven creativity').

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Table 1
Categories of reproduction and their corresponding regulatory approaches.

categories of reproduction	technology involved	costs of reproduction	potential for public distribution/ economic element	regulatory approach
professional reproduction	professional reproduction technology	high	full	regulation by presumption
private reproduction	home reproduction technology	low	limited	regulation by exemption
other exempted uses	compatible	technology- dependent	permitted	
integral reproduction	copy-reliant technology	scarcely any	generally no	non-regulationin general

owners.[6] Third, GenAI does not need to seek authorization from copyright owners but is obliged to pay them, namely, permitted-but-paid use.[7] These viewpoints all presuppose that GenAI's use constitutes 'use' under copyright law, with divergence lying in whether this conduct can be exempted by copyright law and to what extent it applies. However, the preliminary inquiry into whether GenAI's use qualifies as a use under copyright law has received little attention thus far. Nevertheless, without resolving this preliminary question, the basis for discussing how to regulate GenAI's use through copyright law is inadequate. In this paper, we comprehensively explore this preliminary question primarily in the context of U.S. law.

The literature identifies four activities allegedly infringing on the reproduction right. First, GenAI is alleged to acquire copyrighted works through web scraping, which refers to 'copying of web content accomplished using robots or web crawlers.'[8] The digital copies are stored in the training dataset as input data at least until the end of the training process for research outcome verification[9] and are referred to as 'semi-permanent copies'.[10] Second, during the preprocessing stage, GenAI tokenizes copyrighted works ingested as training data by segmenting them into consistent and structured units called tokens.[11] Since 'tokens can be reverse engineered into the original text and thus count as a copy' as suggested by Sag,[12] tokenization can be considered a form of reproduction. Third, GenAI is claimed to reproduce training data to train models,[13] such as making unauthorized intermediate copies[14] or memorizing training data.[15] Although the copyright law community has not conducted a detailed analysis regarding how GenAI uses works during the training phase, this process is believed to inherently 'involve the reproduction of entire works or substantial portions thereof,' according to the U.S. Patent and Trademark Office.[16] Finally, it is claimed that the content generated by GenAI (AI-generated content, AIGC) infringes copyright when it resembles the original works.[17] While there are concerns regarding whether GenAI infringes on the derivative right in this scenario,[18] courts seem reluctant to support such a claim.[19] Furthermore, the derivative right significantly overlaps with the

[6] Jessica L. Gillotte, 'Copyright Infringement in Ai-Generated Artworks' (2020) 53 UC Davis Law Review 2655, 2671 (2020); Benjamin L. W. Sobel, 'Artificial Intelligence's Fair Use Crisis' (2017) 41 Columbia Journal of Law & Arts 45; Ashay Maske, 'Generative AI and IP Infringement' (October 7, 2023) https://ssrn.com/abstract=4595149 accessed 21 July 2024.

[7] Martin Senftleben, 'Generative AI and Author Remuneration' (2023) 54 IIC - International Review of Intellectual Property and Competition Law 1535; Christophe Geiger and Vincenzo Iaia, 'The Forgotten Creator: Towards A Statutory Remuneration Right for Machine Learning of Generative AI' (2024) 52 Computer Law & Security Review 105925; anonymous.

[8] Kathleen C. Riley, 'Data Scraping as a Cause of Action: Limiting Use of the CFAA and Trespass in Online Copying Cases' (2018) 29 Fordham Intellectual Property Media & Entertainment Law Journal 245, 245.

[9] Giorgio Franceschelli and Mirco Musolesi, 'Copyright in Generative Deep Learning' (2022) 4 Data & Policy e17, 2.

[10] Sag, 'Fairness and Fair Use in Generative AI' (n 4) 1893.

[11] Samuelson, 'Generative AI Meets Copyright' (n 3) 159.

[12] Sag, 'Fairness and Fair Use in Generative AI' (n 4) 1893, note 29. Nevertheless, it should be noted that reverse-engineering tokens to recover the exact original text is challenging. Specific features such as punctuation, spacing, and formatting might be lost when converting back and forth. Zhuo Chen, Kui Fan, Shiqi Wang et al., 'Lossy Intermediate Deep Learning Feature Compression and Evaluation' (2019) The 27th ACM International Conference, doi: 10.1145/3343031.3350849. Additionally, different inputs can produce similar token sequences, complicating the exact reconstruction of the original input. While reverse engineering is difficult, it is technically possible under certain conditions. Kai Kugler, Simon Münker, Johannes Höhmann et al., 'Invbert: Reconstructing Text from Contextualized Word Embeddings by Inverting the BERT Ppipeline' (2023) 2(1) Journal of Computational Literary Studies 1, 1.

[13] See Peter Henderson, Xuechen Li & Dan Jurafsky el at., 'Foundation Models and Fair Use' (2023) 24 Journal of Machine Learning Research 1, 2; Lemley & Casey (n 4) 759; Sag, 'Fairness and Fair Use in Generative AI' (n 4) 1894.

[14] Gillotte (n 6) 2671; Lemley & Casey (n 4) 764.

[15] Though Carlini et al. use a rather restricted definition of the notion of 'memorization'—specifically as 'whether diffusion models can be induced to generate near-copies of some training examples when prompted with appropriate instructions. We will describe an approach that can generate images that are close approximations of some training images (especially images that are frequently represented in the training dataset through duplication or other means)' (Nicholas Carlini et al., 'Extracting Training Data from Diffusion Models' (2023) 32nd USENIX Security Symposium (USENIX Security 23) 5253, 5256, note 2)—the notion has been applied beyond the context of generating images or responding to prompts, as long as the generated output closely resembles the input. See Henderson et al. (n 13); Matthew Sag, 'Copyright Safety for Generative AI' (2023) 61(2) Houston Law Review 295. Although prompts are not used during the training stage, the model can still retain specific details from frequently represented examples, leading to outputs that closely resemble the training data. As Henderson et al. note, '[t]he fewer times a model sees an example during training, the less likely it will memorize it' (Henderson et al. (n 13) 22).

[16] United States Patent and Trademark Office, 'Public Views on Artificial Intelligence and Intellectual Property Policy' (October 2020) 24. ('Many comments expressed that the use of copyrighted material to "train" AI may violate the reproduction right of a copyright owner under 17 U.S.C. §106(1), and that this use may or may not be a non-infringing fair use.' (ibid iii); 'When AI algorithms or processes 'learn' their functions by ingesting copyrighted works, reproductions of those works are made in the process as the works are digitized and/or 'read' by the AI algorithms or processes' (ibid 25)).

[17] Sag, 'Fairness and Fair Use in Generative AI' (n 4) 1911; Sobel, 'Artificial Intelligence's Fair Use Crisis' (n 6) 61.

[18] "[E]very output image from the system is derived exclusive from the latent images, which are copies of copyrighted images. For these reasons, every hybrid image is necessarily a derivative work." *Andersen* et al *v. Stability AI Ltd.* et al., Docket No. 3:23-cv-00201 (N.D. Cal. Jan 13, 2023), Court Docket, Complaint ¶ 95.

[19] For example, *Tremblay et al. v. OpenAI, Inc.*, Case Nos. 23-cv-03223-AMO, 8 (the Northern District Court of California argued that though plaintiffs had alleged that every output from GenAI models was an infringing derivative work, they had not provided 'any indication as to what such outputs entail—i.e., whether they are the copyrighted books or copies of the books. That is insufficient to support this cause of action under the DMCA.'); *Kadrey v. Meta Platforms, Inc*, Case No. 23-cv-03417-VC (N.D. Cal. Nov 20, 2023) (the court argued that the allegation of derivative right infringement 'is nonsensical. A derivative work is "a work based upon one or more preexisting works" in any "form in which a work may be recast, transformed, or adapted." 17 U.S.C. § 101. There is no way to understand the LLaMA models themselves as a recasting or adaptation of any of the plaintiffs' books').

reproduction right.[20] Violation of the derivative right necessarily entails unauthorized reproduction of the original expression of a work, with the only difference from violation of the reproduction right being that the derivative creator incorporates his or her own original expression.[21] Consequently, in the last scenario, the alleged infringing action is also reproduction.

However, the view that GenAI involves copying works and consequently constitutes infringement of the reproduction right lacks a comprehensive understanding of the technological process involved in GenAI's use and fails to grasp the correct meaning of the reproduction right. The main contribution of this paper is considering GenAI's use generally as non-regulated use rather than exempted use, infringing use or permitted-but-paid use—an argument with substantial importance.

First, we explicitly posit that copyright-regulated reproduction should encompass not only the technological element but also the economic element. The presence of the technological element determines that a specific behavior qualifies as an act of reproduction, while the existence of the economic element dictates that this act of reproduction is subject to copyright regulation. Highlighting the role of the economic element in regulating a technological act of reproduction is particularly crucial to prevent excessive expansion of copyright in the era dominated by what Sag coined copy-reliant technologies,[22] such as GenAI, where reproduction is essential to their functioning.[23] GenAI does not engage in the reading, comprehension, or enjoyment of copyrighted works, nor does it distribute copies of the works to the public as substitutes for the original work. The sole purpose of GenAI in reproducing these works is to use them as grist to feed algorithms and indices. We also observe the economic rationale for different regulatory approaches to reproduction. The regulation governing the act of reproduction gradually becomes more permissive or even completely vanishes as the potential of its generated copies to undermine copyright owners' economic incentives diminishes, accords with legal provisions or even disappears.

Second, our argument is based on an in-depth technological analysis of GenAI's use of works throughout the data acquisition, preprocessing, training and generation stages. The methods and entities involved in reproduction vary at each stage. Addressing the entire process without distinguishing between these stages, as many commentators have done,[24] leads to unclear and unfocused policy recommendations. By categorizing the process into stages that are easily comprehensible to the industry, this paper helps to enhance communication with industry stakeholders. While GenAI produces copies during the initial two stages, such reproduction falls outside the purview of copyright regulation, as these copies lack the potential of substituting for the original works. During the training and generation stages, only in rare cases does GenAI produce outputs resembling training data and meet the technological element. Since the output has the potential for public distribution and substituting for original works in the generation stage but not the training stage, copyright-regulated reproduction only occurs in the generation stage when AIGC substantially resembles prior works in the training dataset.

This paper is structured as follows. Section 1 clarifies the concept of reproduction within the meaning of copyright law, drawing upon the origin and the purpose of the reproduction right. This reveals that copyright law only regulates reproduction with both technological and economic elements and examines how the economic element of reproduction influences the respective regulatory approach. Section 2 illustrates how GenAI reproduces works in different stages and observes that only during the generation stage, when AIGC is substantially similar to copyrighted works in the training dataset does GenAI's reproduction fall within the scope of copyright regulation. Section 3 addresses the possible criticism that it would result in an unfair situation where GenAI free rides on preexisting works.

2. Reproduction and its regulation under copyright law

To address the question of whether GenAI infringes on the reproduction right, it is imperative to first establish a clear understanding of the meaning of the reproduction right. The name itself implies that the reproduction right—the right to copy—is the earliest form of copyright and holds a central position in copyright law.[25] However, it has frequently been employed without proper comprehension. Section 2.1 highlights that the purpose of the reproduction right is to safeguard copyright owners' economic incentives against unauthorized reproductions that would potentially substitute for original works. This implies that the act regulated by the reproduction right should encompass not only the technological element but also the economic element. Section 2.2 identifies the category of reproduction falling within the scope of copyright law by presumption. Section 2.3 identifies the category of reproduction that falls under copyright law but is exempt from its provisions. Section 2.4 identifies the category of reproduction that lies beyond the scope of copyright law regulation.

2.1. Technological and economic elements of copyright-regulated reproduction

The reproduction right originated from the invention of the Gutenberg printing press.[26] The printing technology has significantly reduced the cost of copying and improved the quality of copies, enabling them to function as viable substitutes for original works. Given that creating works incurs substantially higher costs than copying them, the absence of creators' control over the right of reproduction would result in an influx of unauthorized reproductions into the market, driving down the price of the work to nearly match its reproduction costs. Consequently, creators lose their expectation of recouping their investment in creation and subsequently experience a decline in economic incentives for their creative endeavors. It is the consideration of providing creators with

[20] Some even argue that 'the right to prohibit the making of derivatives is essentially useless and subsumed under the reproduction right.' (Daniel Gervais, 'The Derivative Right, or Why Copyright Law Protects Foxes better than Hedgehogs' (2012) 15 Vanderbilt Journal of Entertainment & Technology Law 785, 799).

[21] The only limited way to independently infringe the derivative right without infringing the reproduction right is in reuse of the original work without making a copy of the work, such as mounting or gluing a picture on a wall tile by using the original image. (ibid 839.) However, in the context of GenAI, if AIGC resembles an original work, it is undoubtedly not a direct use of the original work *per se* but involves the use of a digital reproduction of the original work. The controversy surrounding the question of whether GenAI infringes upon the derivative right ultimately boils down to whether it violates the reproduction right. Therefore, this paper does not delve into the derivative right but concentrates on the reproduction right.

[22] Matthew Sag, 'Copying and Copy-reliant Technology' (2009) 103(4) Northwestern University Law Review 1607, 1608.

[23] Mark A. Lemley, 'The Law and Economics of Internet Norms' (1998) 73 Chicago-Kent Law Review 1257, 1278.

[24] See, for example, Lemley & Casey (n 4); Gillotte (n 6); Sobel, 'Artificial Intelligence's Fair Use Crisis' (n 6); Sag, 'Copyright Safety for Generative AI' (n 15).

[25] Christina Mulligan, 'Copyright without Copying' (2017) 27(2) Cornell Journal of Law and Public Policy 469, 469.

[26] Paul Goldstein, *Copyright's Highway: From Gutenberg to the Celestial Jukebox* (Stanford University Press, Stanford 2003) 21.

economic incentives for creation by allowing them to recoup investments that brings the reproduction right.[27] Therefore, the act of reproduction regulated by the reproduction right does not simply refer to making a copy in a technological sense but making a copy capable of serving as a substitute for the original work in the market.

The technological element determines that a specific behavior qualifies as an act of reproduction. This element's requirement is fulfilled when a particular act results in the generation of a reproduction, whether it is an exact copy or a substantially similar replica, of the original work, regardless of the method employed to generate the reproduction.[28] Nevertheless, it is undeniable that reproduction technology does influence the technological element. Home reproduction technology democratizes the ability to make reproductions and empowers every end user to make reproductions.[29] Digital technology ensures that every reproduction is an exact copy of the original work, thereby alleviating any controversy surrounding whether the similarity between the generated output and the original work is substantial enough to qualify the output as a reproduction and simplifying the fulfillment of the technological element's requirement. Copy-reliant technology driven by reproduction makes every process definable as an act of reproduction in the technological sense.[30]

The economic element dictates that this act of reproduction is subject to copyright regulation. This element refers to the capacity of a reproduction to substitute for the original work in the market and thereby undermines copyright owners' economic incentives. As Liederman claimed, central to the analysis of whether a certain type of reproduction falls within the purview of copyright law hinges upon 'whether it substitutes the original work or has the capacity to do so.'[31] Copyright law solely concerns itself with reproductions 'that can be propertized and thus subject to the economic incentives at the heart of copyright law'.[32] The capacity of a reproduction to serve as a substitute for the original work in the market is closely associated with its potential for public distribution. Given the significance of the economic element in subjecting reproduction to copyright regulation, Pavel argues that copyright law should 'change the proxy for infringement from "copying" to communicating works to the public'.[33] After all, 'the problem is not one of reproduction, but rather of distribution'.[34]

The economic element is also influenced by technology. Prior to home reproduction technology, reproduction incurred significant costs. To recover these costs, reproducers had to distribute large quantities of reproductions publicly, which had the potential to substitute original works and undermine copyright owners' incentives.[35] Consequently, reproduction carried out by professional reproduction technology possesses the economic element. The emergence of home reproduction technology has reduced the costs of reproduction and the necessity for the public distribution of reproductions, thereby mitigating negative impacts on copyright owners' economic incentives and limiting the economic element of reproduction. Copy-reliant technology integrates reproduction as an inherent function without incurring additional expenses. These reproductions are made solely to facilitate machine processing without any potential for public distribution, thus lacking the economic element generally. For example, GenAI reproduces works as the raw material to train its algorithms rather than making publicly distributable reproductions to substitute for the original works.[36]

2.2. Regulation by presumption: the hidden economic element

Despite the importance of the economic element for an act of reproduction to fall under the purview of copyright law regulation, the Berne Convention and many national copyright laws explicitly incorporate the technological rather than the economic element's requirement into the definition of reproduction. For example, Article 9(1) of the Berne Convention states that the reproduction right regulates reproduction 'in any manner or form' without mentioning the economic impact of reproduction. This approach is followed by the German copyright law, which provides that the reproduction right 'means the right to produce copies of the work … regardless of by which means of procedure or in which quantity they are made'.[37] The Chinese copyright law lists various means of reproduction brought about by existing technologies, such as 'printing, photocopying, lithographing, making a sound recording or video recording, duplicating a recording, duplicating a photographic work, and digitalization', with a catch-all provision of reproduction 'by any other means'.[38] The U.S. copyright law defines reproduction as 'to reproduce the copyrighted work in copies or phonorecords'[39] where 'copies' are defined as material objects 'in which a work is fixed by any method now known or later developed, and from which the work can be perceived, reproduced, or otherwise communicated, either directly or with the aid of a machine or device.'[40] This implies that any yet undiscovered method capable of fixing a work to enable its perception through an unknown machine or device, regardless of its potential for public distribution, would fall within the scope of the reproduction right.

The emphasis on the technological element of reproduction, accompanied by the open approach towards the means of reproduction, might be attributed to the strong association between technological advancements and the reproduction right which is considered 'the first entitlement to face the challenges raised by technological developments'.[41] New technologies introduce novel means of reproduction. Every time a new technology emerges, copyright owners strive for control over the new means of reproduction. The expansion of the reproduction right reached its pinnacle with its provision under the Berne Convention stating that reproduction 'in any manner or form' is regulated. At the time of legislation, this broad prescription might simply be a proactive measure taken by policymakers to safeguard copyright owners' interests against potential infringement from

[27] 'Copyright is based on the belief that by granting authors the exclusive rights to reproduce their works, they are given an incentive to create, and that "encouragement of individual effort by personal gain is the best way to advance public welfare through the talents of authors and inventors in 'Science and the useful Arts.'" *Sony Corp. of America v. Universal City Studios, Inc.*, 464 U.S. 417, 477 (1984).

[28] Joseph P. Fishman, 'The Copy Process' (2016) 91 New York University Law Review 855, 855.

[29] Alison Powell, 'Democratizing Production through Open Source Knowledge: From Open Software to Open Hardware' (2012) 34(6) Media, Culture & Society 691.

[30] Sag, 'Copying and Copy-reliant Technology' (n 22).

[31] Joshua C. Liederman, 'Changing the Channel: The Copyright Fixation Debate' (2009) 36 Rutgers Computer & Technology Law Journal 289, 317.

[32] Laura A. Heymann, 'How to Write a Life: Some Thoughts on Fixation and the Copyright/Privacy Divide' (2009) 51 William & Mary Law Review 826, 849.

[33] Ashley M. Pavel, 'Reforming the Reproduction Right: The Case for Personal Use Copies' (2009) 24(4) Berkeley Technology Law Journal 1615, 1615.

[34] Mulligan (n 25) 480.

[35] Jane C. Ginsburg, 'Copyright and Control Over New Technologies of Dissemination' (2001) 101 Columbia Law Review 1613.

[36] Michael D. Murray, 'Generative AI Art: Copyright Infringement and Fair Use' (2023) 26(2) SMU Science and Technology Law Review 259, 276.

[37] Section 16 of Act on Copyright and Related Rights of Germany (Copyright Act, as amended up to Act of June 23, 2021)

[38] Article 10 of Copyright Law of the People's Republic of China.

[39] 17 U.S.C. § 106.

[40] 17 U.S.C. § 101.

[41] Caterina Sganga, 'The Right of Reproduction' in Eleonora Rosati (ed) *The Routledge Handbook of EU Copyright Law* (Routledge, 2021) 123.

emerging technologies.[42] However, in an era dominated by copy-reliant technology, which necessitates routine and indiscriminate copying of works for machine processing, thereby making reproduction technically ubiquitous,[43] such a prescription renders any action performed by digital machines or devices subject to the regulation under copyright law—making the reproduction right 'perhaps the least suitable for the digital environment due to its technical-functional nature.'[44]

The hiding of the economic element is influenced by the presumption that reproduction inherently possesses the economic element. When the provision for the reproduction right was formulated, copying necessitated substantial capital-intensive equipment, whether 'a printing press or a garage full of VCRs'.[45] To recover the costs of copying, copiers would engage in large-scale sales of copies. Given that copies could be openly distributed, they would potentially serve as substitutes for original works and undermine the economic incentives for copyright owners; thus, at that time, reproduction possessed both technological and economic elements. This is why although the legislation clauses do not literally mention the economic element of reproduction, copyright law regulates the public distribution of reproductions rather than the act of reproduction itself. In other words, it is still the economic element resulting from the public distribution of copies that subjects an act of reproduction to copyright regulation.

2.3. Regulation by exemption: the limited or permitted economic element

The emergence of household photocopying machines has weakened the economic element of reproduction. These machines have significantly diminished the marginal cost of copying, reducing the need for the public distribution of copies. As private copies do not enter the market to compete with the original work, although they physically reproduce the work, these copies, if in small scale, have little potential to interfere with the normal exploitation of the original work and undermine the economic incentives for copyright owners. Consequently, private copying is exempt across countries on the condition that it is confined to personal and non-commercial use.[46] While a single private copy may not adversely impact the market for the original work, an extensive proliferation of private copies would indeed do so.[47] Thus, copyright exceptions for private copying are often accompanied by levy schemes to compensate copyright owners to limit the economic element of private copying.[48] The limited economic element of private reproduction forms the basis for its exemption from copyright regulation.

Private use is one example of exempted uses under copyright law. In the U.S., copyright exemptions are typically governed by the 'fair use' doctrine. Although European countries do not have a notion of fair use, they also incorporate various copyright exceptions. Both the U.S. and European countries exempt uses for purposes such as private use, criticism, news reporting, and teaching.[49] The justifications for exempting private use and other types of exempted uses differ.[50] The exemption of private use is primarily justified by the disproportionately high transaction costs of obtaining permission for reproduction compared with the potential benefits, as well as the minimal harm it poses to copyright owners' economic incentives.[51] In essence, the economic element of private use is limited.

Conversely, other types of exempted uses, whether referred to as fair use in the U.S. or as copyright exceptions in European countries,[52] are justified by their contributions to a broader public interest, despite their impact on the copyright market.[53] For example, reproducing one chapter of a textbook and distributing the copies to the class eliminates

[42] Martin Senftleben, 'Compliance of National TDM Rules with International Copyright Law: An Overrated Nonissue?' (2022) 53 IIC-International Review of Intellectual Property and Competition Law 1477, 1499.

[43] Sag, 'Copying and Copy-reliant Technology' (n 22) 1617.

[44] Taina Pihlajarinne, 'Should We Bury the Concept of Reproduction – Towards Principle-Based Assessment in Copyright Law?' (2017) 48 IIC - International Review of Intellectual Property and Competition Law 953, 953.

[45] Mulligan (n 25) 480.

[46] Section 53 of Act on Copyright and Related Rights of Germany; Article 24 of Copyright Law of China; Section 43C of Copyright Act 1968 (consolidated as of July 1, 2022) of Australia. In the U.S., although private use is not specifically listed in the fair use clause (17 U.S. Code § 107), it is generally considered a type of fair use. As Gordon noted, 'the "personal," "individual" nature of copying has been held relevant to fair use.' Wendy J. Gordon, 'Fair Use as Market Failure: Structural and Economic Analysis of the Betamax Case and Its Predecessors' (1983) Journal of the Copyright Society of the USA 253, 289. Also see L. Ray Patterson, 'Understanding Fair Use' (1992) 55(2) Law and Contemporary Problems 249, 261; Neil Weinstock Netanel, 'Making Sense of Fair Use' (2011) 15 Lewis & Clark Law Review 715, 771.

[47] Glynn S. Lunney Jr, 'The Death of Copyright: Digital Technology, Private Copying, and the Digital Millennium Copyright Act' (2001) 87 Virginia Law Review 813, 818; Diane Leenheer Zimmerman, 'Modern Technology, Leaky Copyrights, and Claims of Harm: Insights from the Curious History of Photocopying' (2013) 61 Journal of the Copyright Society of the U.S.A. 1, 10.

[48] Martin Krestschmer, *Private Copying and Fair Compensation: An Empirical Study of Copyright Levies in Europe* (Intellectual Property Office, 2011). In the U.S., the 1992 Audio Home Recording Act (AHRA) requires manufacturers, importers and distributors of digital audio recording devices to remunerate the record labels with a predetermined amount: a statutory levy of 3% of the 'transfer price' of each digital audio tape or 2% of the 'transfer price' of each digital audio recording device. 17 U.S. Code § 1004(a)(1) & (b).

[49] The fair use clause in the U.S. (17 U.S. Code § 107) explicitly exempts uses for purposes such as criticism, comment, news reporting and teaching. Although private use is not specifically listed, it is generally considered a type of fair use. See *supra* note 46. In Europe, Germany exempts uses for purposes such as 'reporting on current events' (Section 50), 'caricature, parody and pastiche' (Section 51a), 'private and other personal uses' (Section 53), and 'teaching in educational establishments' (Section 60a). France exempts uses for purposes such as 'parody, pastiche and caricature', 'strictly private use', and 'exclusive illustrating purposes in the context of teaching' (Article L122-5). The Netherlands exempts uses for purposes such as 'quotation' (Section 15a), 'illustration for teaching' (Section 16), and 'private practice' (Section 16b).

[50] As Gordon noted, private use constitutes an 'excuse' for a fair-use decision due to market malfunction. In cases of other types of fair use such as news reporting and teaching, the market functions but has inherent limitations in internalizing non-economic values. These non-economic interests serve as a 'justification' for fair use. Wendy J. Gordon, 'Excuse and Justification in the Law of Fair Use: Transaction Costs have always been Part of the Story' (2002) 50 Journal of Copyright Society of USA 149, 153. Notably, according to certain perspectives, safeguarding privacy and fostering innovation in new technologies also justify exempting private copying. See, for example, Stavroula Karapapa, *Private Copying* (Routledge, 2012) 2 ('Private use…finds protection under privacy considerations which come into play at the stage of copyright enforcement'); Fred Von Lohmann, 'Fair Use as Innovation Policy' (2008) 23 Berkeley Technology Law Journal 829, 839 ('the fair use doctrine sets private copying beyond a copyright owner's reach to encourage the development of new technologies'). Similarly, other types of exempted uses may also have alternative rationales. However, for simplicity and clarity of our argument, we differentiate private use from other exempted uses based on the economic element, similar to Gordon's differentiation of various exempted uses according to the presence of market malfunction or inherent limitations.

[51] William M. Landes and Richard A. Posner, *The Economic Structure of Intellectual Property Law* (Harvard University Press 2003) 117.

[52] The U.K. follows the fair dealing tradition, distinct from the copyright exception system commonly observed in many European countries.

[53] Landes and Posner (n 51) 122 (identifying a category of fair use that 'involves some harm to the copyright holder in lost revenues, but the harm is more than offset by the sum of the benefits to others').

the necessity for every student to purchase the textbook and thus affects the market of the original work, but it promotes and facilitates access to education and is permitted by copyright law.[54] Thus, the economic element of other types of exempted uses, if present, is generally permitted by copyright law. Since the economic element is closely linked to the potential for public distribution—related to the cost of reproduction and reproduction technology—other exempted uses are less tied to specific technology and can be performed through professional reproduction technology, home reproduction technology and copy-reliant technology. The potential for public distribution varies depending on the employed technology, while copyright law permits such potential.

The consideration of the economic element in determining whether a reproduction falls under copyright regulation does not override the tests for determining exempted uses, either the last two steps under the Berne three-step test[55] or the fourth factor under the U.S. four-factor test.[56] Copyright law only regulates acts of reproduction that possess the economic element, namely those acts capable of generating copies that have the potential to damage copyright owners' economic incentives. If an act of reproduction has a limited or permitted economic element, it may qualify for exemption. If it completely lacks the economic element, it is not subject to copyright regulation and does not require an assessment for exemption. Therefore, incorporating the notion of economic element elucidates the rationale for distinct regulatory approaches to different categories of reproduction.

2.4. Non-regulation: lack of the economic element

Copy-reliant technology, which operates through the routine reproduction of works, has made it possible to completely separate the economic element from reproduction. A typical example is that, for a computer to access and perceive a work, it needs to automatically copy the work into the random access memory (RAM) as a necessary part of its routine operation, which is referred to as temporary reproduction.[57] In contrast to private copying, which aims to retain copies with potential for public distribution (though such potential is subject to restrictions), temporary reproduction aims to make a work perceptible rather than obtain copies. The generation of temporary copies is simply part of the process of accessing or using the original work. The copies are temporarily stored in RAM during the immediate computing session and are typically not saved or accessible outside it.[58] They are erased upon power-off or overwritten by new data. Temporary copies cannot be accessed or exploited separately from the original work, let alone compete with it or be substitutes in its market. Therefore, temporary copies would not have any economic implications on the market for the original work and should stay outside the scope of copyright regulation.

This argument has been substantiated by copyright legislation. Although the reproduction right provision under the Berne Convention encompasses reproduction 'in any manner or form',[59] which has sparked significant controversy over whether temporary reproduction should be regulated,[60] it is noteworthy that this provision was adopted during the 1967 Stockholm Conference, wherein policymakers had no temporary reproduction in mind.[61] The WIPO Copyright Treaty ('WCT') ultimately rejected such a broad provision of the reproduction right due to the strong opposition from the 1996 WIPO Diplomatic Conference on Certain Copyright and Neighboring Rights Questions ('the 1996 diplomatic conference').[62] The statement regarding the reproduction right under the WCT can only be found in the Agreed statement concerning Article 1(4), which argues that 'the reproduction right … fully apply in the digital environment', and 'the storage of a protected work in digital form in an electronic medium constitutes a reproduction within the meaning of Article 9 of the Berne Convention.' However, it should be noted that this Agreed statement has not been adopted by consensus, with 51 votes in favor, 5 against, and 30 abstentions; thus, its impact and interpretative authority are limited.[63]

Some argue that temporary reproduction falls outside copyright regulation due to its technological nature. For example, during the 1996 diplomatic conference, the Danish delegate contended that 'temporary reproduction made for the sole purpose of making a work perceptible, or reproduction of a purely transient or incidental character, as part of a technical process, did not constitute reproduction within the meaning of Article 9(1) of the Berne Convention.'[64] This view resonated with Norway,[65] Australia,[66] Switzerland[67] and the African Group.[68] However, we argue that it is the economic nature rather than the technological nature of temporary reproduction that drives it beyond copyright regulation.

First, it is determined by the inherent challenges in delineating the scope of temporary reproduction from a technological standpoint. The National Commission on New Technology Uses has affirmed that distinguishing transient and incidental reproduction from non-transient or non-incidental reproduction poses significant difficulties technologically. Attempting to determine whether a 'copy' was temporary through a temporal threshold would be 'futile' owing to the rapid advancement of reproduction technology.[69] This is why courts have declined to apply

[54] *Cambridge University Press v. Patton*, 769 F.3d 1232, 1250 (11th Cir. 2014); *Princeton University Press v. Michigan Document Services, Inc*. 99 F.3d 1381, 1404 (6th Cir. 1996).

[55] '[S]uch reproduction does not conflict with a normal exploitation of the work and does not unreasonably prejudice the legitimate interests of the author.' Article 9(2) of Berne Convention.

[56] '[T]he effect of the use upon the potential market for or value of the copyrighted work' 17 U.S. Code § 107.

[57] R. Anthony Reese, 'The Public Display Right: The Copyright Act's Neglected Solution to the Controversy over RAM Copies' (2001) 2001(1) University of Illinois Law Review 83, 84.

[58] Ibid 84.

[59] Article 9(1) of the Berne Convention.

[60] For example, Pelin Karaaslan, 'Temporary Reproduction of Intellectual and Artistic Works due to Technical Requirement' (2022) 2022(1) Journal of Intellectual Property Law 29, 29; Aaron Perzanowski, 'Fixing Ram Copies' (2010) 104(3) Northeastern University Law Review 1067, 1069; Raymond Shih Ray Ku, 'The Creative Destruction of Copyright: Napster and the New Economics of Digital Technology' (2002) 69(1) The University of Chicago Law Review 263, 265; Reese (n 63) 86.

[61] Senftleben (n 42) 1498.

[62] WIPO, *Records of the Diplomatic Conference on Certain Copyright and Neighboring Rights Questions, Geneva 1996* (WIPO 1999), Vol. II, 764, paras. 877–881; Senftleben (n 42) 1489.

[63] Senftleben (n 42) 1489.

[64] WIPO (n 62) 669, para. 256.

[65] WIPO, *Records of the Diplomatic Conference on Certain Copyright and Neighboring Rights Questions, Geneva 1996,* (WIPO, 1999), Vol. I, 407, Document CRNR/DC/22.

[66] Ibid 433–434, Document CRNR/DC/53. An Australian delegate proposed that 'paragraph (1) shall not apply to indirect or temporary reproductions that have the sole purpose of making a work perceptible or which are of a purely transient or incidental character as part of a technical process.'

[67] WIPO (n 62) 673, para. 281.

[68] WIPO (n 65) 445, Document CRNR/DC/56.

[69] National Commission on New Technological Uses of Copyrighted Works, Final Report (1978) 22 https://digital-law-online.info/CONTU/PDF/Chapter3.pdf accessed 21 July 2024.

the transitory durational requirement to determine whether a copy is transitory or not[70] or even considered reproductions that last for more than a 'transitory duration' as unfixed, transitory copies.[71]

Second, it is determined by the economic logic underlying the establishment of legal rights. Regardless of the feasibility of distinguishing between temporary and non-temporary reproduction from a technological perspective, the emergence of property rights is driven by economic forces.[72] Legal intervention, being the most costly regulatory measure, is not warranted for issues lacking economic significance.[73] The technological nature of temporary reproduction ensures that this type of reproduction does not undermine copyright owners' economic incentives.[74] The lack of the economic element excludes temporary reproduction from copyright law's purview. Just as Korea stated at the 1996 diplomatic conference, 'reproductions that were not relevant in economic terms should not be considered as reproductions'.[75] Sweden confirmed that temporary reproductions 'had no economic significance' and 'were not really reproductions'.[76] This is why even proponents advocating for the inclusion of temporary reproduction within copyright law acknowledge that this proposition stems from the fact that 'the economic value of certain types of temporary reproduction has gained increased relevance to the sustaining of copyright owners' incentives',[77] while also emphasizing that 'temporary copies that have no such incentive-producing function' should be excluded from copyright regulation.[78]

Focusing on the economic element also gains support from an official perspective. For example, during the 1982 deliberations of the Second WIPO/UNESCO Committee of Governmental Experts on Copyright Problems Arising from the Use of Computers for Access to or the Creation of Works, the Committee clarified that an act constituting reproduction within the meaning of the reproduction right may 'involve fixation of works in a form sufficiently stable to permit their communication to an individual.'[79] The committee explicitly stated that the requirement for stable fixation serves the purpose of ensuring that the works can communicate to the public. This view is also in harmony with the U.S. Copyright Office's report, which posits that for a 'reproduction' to fall under copyright regulation, what matters is its capacity to be perceived or reproduced to substitute for the original work but not the specific period it is held or stored.[80]

Temporary reproduction exemplifies the category of non-regulated reproduction. Reproduction facilitated by copy-reliant technology, which inherently integrates reproduction as a functional aspect without incurring additional costs solely for machine processing purposes, is generally devoid of any potential for public distribution and falls within this category owing to the absence of the economic element.

The spectrum of reproduction regulated by presumption, reproduction regulated by exemption and non-regulated reproduction and the underlying economic rationales are illustrated in Table 1.

3. How GenAI uses works

This section provides a comprehensive analysis of the entire process of GenAI's use of works. During the data acquisition and preprocessing stages, GenAI reproduces copyrighted works ingested as training data; however, these reproductions lack potential for public distribution and substituting for the original works. During the training and generation stages, only in rare cases when the output data resemble the input data does GenAI reproduce works and encompass the technological element. Nevertheless, only in the generation stage does the output have the potential to undermine copyright owners' incentives. Therefore, GenAI's reproduction falls within the scope of copyright law only during the generation stage when AIGC exhibits substantial similarity to works in the training dataset.

3.1. Data acquisition stage

The initial step is to acquire raw data, a substantial portion of which comprises copyrightable works, such as text, images, audio and video. Most of the works are publicly available on the Internet and have been extracted by GenAI providers via web scraping tools such as BeautifulSoup and Scrapy.[81] Web scraping involves the digital reproduction of web content[82] and possesses the technological element. Prior to GenAI, web scrapers may display the extracted content online for public distribution, often substituting for original works and impacting their market.[83] This is why web scraping is considered to infringe the reproduction right as suggested by some.[84] Nevertheless, GenAI's web scraping does not, and does not intend to, publicly distribute or provide access to the scraped works. Instead, it aims to extract raw data, followed by cleaning[85] and transforming the data into a format suitable for training GenAI and then storing the scraped and structured data for preprocessing.[86] Given that GenAI's web scraping does not generate output capable of substituting or potentially substituting for the original works in the market, it lacks the economic element and thus falls outside the scope of copyright regulation.

Some scholars have argued that the copies generated through web

[70] In *MAl Systems Corp. v. Peak Computer*, 991 F.2d 511 (9th Cir. 1993), the court 'does not discuss the "transitory duration" prong of the fixation test'; in *Marobie-FL, Inc. v. Nat'l Ass'n of Fire Equip. Distrib.*, 983 F. Supp. 1167, 1177-78 (N.D. Ill. 1997), the court determines that a work momentarily stored in a RAM buffer and then immediately transmitted from a host computer to the Internet constituted a fixed copy but 'do[es] not apply the durational requirement with the Copyright Act'.

[71] See, for example, CoStar Group, Inc. v. Loopnet, Inc., 373 F.3d 544, 551 (4th Cir. 2004).

[72] See Harold Demsetz, 'Towards a Theory of Property Rights' (1967) 57(2) The American Economic Review 347, 347.

[73] Robert C. Ellickson, *Order without Law: How Neighbors Settle Disputes* (Harvard University Press, 1994) 214.

[74] Haochen Sun, 'Reconstructing Reproduction Right Protection in China' (2005-2006) 53(1) Journal of the Copyright Society of the U.S.A 223, 268, note 194.

[75] WIPO (n 65) 628, para. 501.

[76] WIPO (n 62) 670, para. 263.

[77] Sun (n 74) 267.

[78] Ibid 268, note 194.

[79] Berne Union, 'Second Committee of Governmental Experts on Copyright Problems Arising from the Use of Computers for Access to or the Creation of Works (Paris, June 7 to 11, 1982)' (1982) 9 Monthly Review of the World Intellectual Property Organization Copyright 239, 243, para. 33.

[80] United States Copyright Office, DMCA Section 104 Report (2001) 110 https://www.copyright.gov/reports/studies/dmca/sec-104-report-vol-1.pdf accessed 21 July 2024.

[81] Talin Natali Ghazarian, 'We Built This City: Generative AI, Copyright, and the Built Environment' (2023-2024) 51 Fordham Urban Law Journal 1227, 1228; Helberger, 'FutureNewsCorp, or How the AI Act Changed the Future of News' (2024) 52 Computer Law & Security Review 105915.

[82] Riley (n 8) 245; Benjamin L. W. Sobel, 'A New Common Law of Web Scraping' (2021) 25 Lewis & Clark Law Review147, 161.

[83] For example, the Facebook Crawler scans the HTML source of any page shared on Facebook or Messenger to collect data, including the page title, meta description, and a thumbnail image, to create a preview that is publicly distributable and available online. See Datadome, How Facebook Was Used as a Proxy by Web Scraping Bots, https://datadome.co/threat-research/how-facebook-was-used-as-a-proxy-by-web-scraping-bots/ accessed 21 July 2024.

[84] Sobel, 'A New Common Law of Web Scraping' (n 82) 169-170.

[85] Xu Chu, Ihab F. Ilyas, Sanjay Krishnan & Jiannan Wang, 'Data Cleaning: Overview and Emerging Challenges' (2016) Proceedings of the 2016 international conference on management of data 2201, 2202-2203.

[86] See Clive Davies, 'Generative AI and IT Outsourcing' (2024) 17 International In-House Counsel Journal 8837, 8842; Jon M. Garon, 'The Revolution Will Be Digitized: Generative AI, Synthetic Media, and the Medium of Disruption' (2023) 20 Ohio State Technology Law Journal 139, 199.

scraping are stored in the training dataset at least until the training process is complete,[87] thereby excluding the copies from the temporary reproduction exception[88] and triggering infringement upon the reproduction right.[89] However, whether the copies are temporarily or permanently stored is immaterial; what matters is whether they have the economic element. Since GenAI stores these copies in the training set without potential for public dissemination, they remain outside the purview of copyright regulation. Occasionally, a training dataset itself may be distributed. However, since it includes preprocessed data in mathematical representations, its distribution is typically intended for training GenAI models rather than substituting the original works contained in the dataset. Those who obtain such a dataset use it as an internal raw material for training their models rather than distributing reproductions in the market to replace the original works, and thus such use still lacks the economic element.

Even in rare cases where a training dataset is distributed to substitute for original works and undermines copyright owners' economic incentives, copyright-regulated reproduction occurs during the distribution stage rather than the preprocessing stage, with the infringer being the distributor of the training dataset rather than the data acquirer. Notably, such infringement is unrelated to GenAI's technological process and can potentially occur in the distribution of any object. Similarly, in rare cases where the AIGC is similar to the training dataset and qualifies as a reproduction with potential for public distribution, possessing both technological and economic elements, infringement may occur during the generation stage, as discussed in Section 2.4, rather than during the data acquisition stage, with the infringer being the GenAI model creator rather than the data acquirer.[90]

In summary, GenAI does not engage in copyright-regulated reproduction within the data acquisition stage.

3.2. Preprocessing stage

The acquisition of raw data is followed by preprocessing. A copyright-relevant activity in preprocessing is tokenization.[91] GenAI tokenizes works, especially textual works, into individual words or subword units. Each token is represented by a vector in a high-dimensional embedding space, such as Word2Vec or GloVe embeddings.[92] These vectors capture semantic and contextual information about the corresponding tokens and facilitate mathematical operations such as similarity calculations and transformations.[93] This process serves as a prerequisite for training the GenAI model. Both raw data and preprocessed data in the form of tokens and vectors comprise the training dataset.[94]

Sag argued that '[t]he tokens can be reverse engineered into the original text and thus count as a copy',[95] suggesting that tokenization meets the technological element of reproduction. However, unlike prior reproduction technologies that reproduce works in their original format and thus pose the risk of producing substitutes for the original works, GenAI reproduces works in an entirely distinct format, namely, tokens, which are readable for machines but incomprehensible to humans. Tokens generally do not contain enough original expression and are unsuitable for human consumption, rendering them inadequate substitutes for the original works. Furthermore, tokens are stored in the training dataset without the potential for public distribution and thus lack the economic element. As a result, tokenization cannot be regarded as copyright-regulated reproduction.

While there may be a possibility of GenAI models reproducing tokens and presenting them in the form of AIGC with the potential for public distribution, the question of whether copyright-regulated reproduction occurs should be addressed in the generation stage rather than the preprocessing stage, with the potential infringer being the GenAI model creator rather than the data preprocessor.[96]

In summary, GenAI does not engage in copyright-regulated reproduction within the preprocessing stage.

3.3. Training stage

There appears to be a consensus within the academic community regarding the infringement of the reproduction right arising from the training process.[97] Even proponents who argue that GenAI's use generally constitutes fair use acknowledge instances of infringement during specific training cases.[98] Ironically, little attention has been given to which reproduction actions are involved in the training process from a technological perspective.

The essence of the training process lies in adjusting parameters, which consist of numerical values that control how a GenAI model learns and generates content. The objective is to minimize the difference between the generated output and the desired output, quantified by a loss function.[99] Parameter optimization relies primarily on iteratively exposing the GenAI model to training data.[100] The model encodes the training data into a mathematical representation to capture the statistical patterns, word relationships, grammar rules, structural variations, and diversity within the underlying data distribution.[101] Subsequently, the model decodes the mathematical representation and generates new, previously unseen data that are consistent with the learned distribution.[102] On the basis of the feedback regarding the disparity between the generated output and the desired output, subsequent rounds of training are conducted where the model is once again exposed to training data.[103] Some argue that during the training process, the model

[87] Franceschelli & Musolesi (n 9); Sag, 'Copyright Safety for Generative AI' (n 15) 314.

[88] Theodoros Chiou, 'Copyright Lessons on Machine Learning: What Impact on Algorithmic Art' (2019) 10 Journal of Intellectual Property, Information Technology and Electronic Commerce Law 398, 407; Isaac Sandiumenge, 'Copyright Implications on the Use of Generative AI' (2023) 12 https://ssrn.com/abstract=4531912 accessed 21 July 2024.

[89] Sag, 'Fairness and Fair Use in Generative AI' (n 4) 1893.

[90] Murray (n 36) 285.

[91] Samuelson (n 3) 159.

[92] E. Hindocha, V. Yazhiny, A. Arunkumar & P. Boobalan, 'Short-text Semantic Similarity Using GloVe Word Embedding' (2019) 6(4) International Research Journal of Engineering and Technology 553, 553.

[93] Raj Bridgelall, 'Unraveling the Mysteries of AI Chatbots' (2024) 57 Artificial Intelligence Review 89.

[94] TensorFlow, 'Data Preprocessing for ML: Options and Recommendations' https://www.tensorflow.org/tfx/guide/tft_bestpractices accessed 21 July 2024.

[95] Sag, 'Fairness and Fair Use in Generative AI' (n 4) 1893, note 29. Nevertheless, as noted in *supra* note 12, reverse-engineering tokens to accurately retrieve the original text is challenging, albeit not impossible.

[96] Murray (n 36) 285.

[97] *Supra* notes 13-16 and the accompanying text.

[98] See, for example, Sag, 'Copyright Safety for Generative AI' (n 15) 295-296; Lemley & Casey (n 4) 748, 750.

[99] Alaa Abu-Srhan, Mohammad A. M. Abushariah & Omar S. Al-Kadi, 'The Effect of Loss Function on Conditional Generative Adversarial Networks' (2022) 34(9) Journal of King Saud University-Computer and Information Sciences 6977, 6977.

[100] Ken Huang, Ben Goertzel, Daniel Wu and Anita Xie, 'GenAI Model Security' in Huang, K., Wang, Y., Goertzel, B., Li, Y., Wright, S., Ponnapalli, J. (eds) *Generative AI Security. Future of Business and Finance* (Springer: Cham 2024) 166.

[101] Min Oh & Liqing Zhang, 'Generalizing Predictions to Unseen Sequencing Profiles via Deep Generative Models' (2022) 12(1) Scientific Reports, Article number 7151.

[102] Ibid.

[103] Ibid.

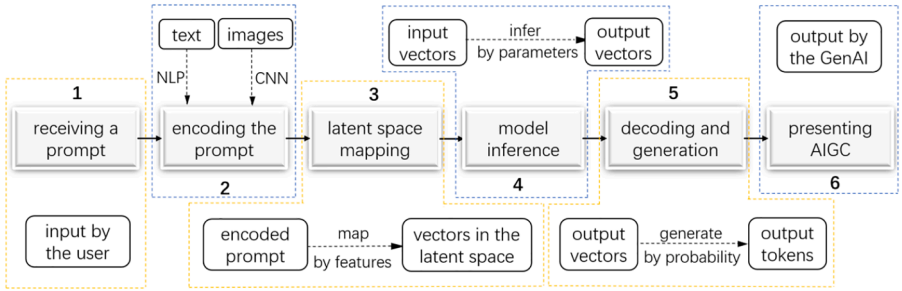

Fig. 1. The generation stage (with solid arrows denoting the direction of flow and dotted arrows signifying the specific operational methods).

'repeatedly encode[s] and decode[s] training data or elements thereof',[104] thereby 'making the intermediate copies of the data'.[105] Nevertheless, it should be noted that the data generated through the model's decoding are not necessarily similar to the encoded data[106] and therefore may not be considered copies. In fact, the purpose of model training is to generate new, previously unseen data with statistical properties consistent with those of the input data rather than generating copies of the input data itself.[107] Therefore, GenAI models normally do not reproduce input data and lack the technological element.

Some argue that in certain instances, GenAI models may 'memorize'[108] the input data and reproduce it in the output, a phenomenon known as overfitting.[109] The true reason for overfitting is that models 'reconstruct idiosyncrasies of input data instead of reflecting underlying trends about those data'.[110] Technically speaking, the models capture noise, which refers to random variability or errors present in the input data, rather than meaningful patterns or signals relevant to the task at hand.[111] This usually happens when the complexity of the model surpasses the size and diversity of the training dataset, when the training duration is excessive, or when the preprocessing of training data is insufficient.[112] Imagine training a language model with a substantial number of parameters capable of capturing complex language patterns to generate authentic product reviews on the basis of a dataset comprising only 200 product reviews of smartphones from a single brand. Given the limited size and specificity of this dataset, the model may encounter challenges in generating reviews for diverse product categories or expressing varied opinions not seen in the training data. Consequently, the generated reviews might exhibit repetitive patterns or verbatim reuse of phrases from the training examples, reflecting the model's inclination toward memorization rather than generalization.[113] In this scenario, overfitting occurs because the model's complexity enables it to accurately learn and reproduce specific nuances and patterns present in the small training dataset; however, this comes at the expense of its ability to generalize and generate diverse outputs beyond what was seen in the training examples.[114] Nevertheless, even if the output generated by the model resembles a work in the training dataset, during the training stage, the output is limited to the test data without the potential for public distribution.

In summary, GenAI does not engage in copyright-regulated reproduction within the training stage.

3.4. Generation stage

The process by which GenAI generates output, i.e., AIGC, can be described as follows (see Fig. 1). First, the user provides a prompt, namely, a specific instruction provided to the GenAI model to guide its generation of output.[115] Second, the model encodes the prompt into a structured and machine-readable format through various techniques, such as natural language processing (NLP) for text and convolutional neural networks (CNNs) for images, to extract features from the prompt.[116] Third, the processed and encoded prompt is mapped to a specific vector representation within the latent space.[117] The latent space is conceptualized as a compressed and structured representation of vectors grouping data with similar features together,[118] allowing the model to generalize and generate new data that adheres to the underlying distribution of the prior data.[119] Fourth, based on the input vectors and optimized parameters learned during the training stage, the model infers output vectors.[120] Fifth, the model decodes the output vectors by transforming them into a meaningful sequence of tokens, where each token is generated probabilistically based on its likelihood, given the preceding tokens.[121] Finally, the model presents the output text or sequence on the basis of these decoded tokens.[122]

The above generation process can generally be described as 'encoding – mapping – inference – decoding', which ensures that GenAI models

[104] Sobel, 'Artificial Intelligence's Fair Use Crisis' (n 6) 62, note 97.
[105] Gillotte (n 6) 2671, note 83.
[106] Murray (n 36) 294.
[107] Matteo Cinselli, Luca Pascarella & Gabriele Bavota, 'To What Extent do Deep Learning-based Code Recommenders Generate Predictions by Cloning Code from the Training Set?' (2022) Proceedings of the 19th International Conference on Mining Software Repositories 167, 169.
[108] Sag, 'Copyright Safety for Generative AI' (n 15) 295.
[109] Xue Ying, 'An Overview of Overfitting and its Solutions' (2019) 1168 Journal of Physics: Conference Series Article number 022022.
[110] Sobel, 'Artificial Intelligence's Fair Use Crisis' (n 6) 64.
[111] Ibid.
[112] ENCORD, 'Overfitting in Machine Learning: How to Detect and Avoid Overfitting in Computer Vision?' https://encord.com/blog/overfitting-in-machine-learning/ accessed 21 July 2024.
[113] Wei Jiaheng et al., 'Memorization in Deep Learning: A Survey' (2024) 18(9) Journal of Latex class files 1.
[114] Omid Poursaeed et al., 'Robustness and Generalization via Generative Adversarial Training' (2021) Proceedings of the IEEE/CVF International Conference on Computer Vision 15711, 15711.
[115] Sag, 'Copyright Safety for Generative AI' (n 15) 300.
[116] Liqi Yan, Cheng Han, Zenglin Xu, Dongfang Liu & Qifan Wang, 'Prompt Learns Prompt: Exploring Knowledge-aware Generative Prompt Collaboration for Video Captioning' (2023) Proceedings of international joint conference on artificial intelligence (IJCAI) 1622, 1622 (2.
[117] Paulino Cristovao et al, 'Generating In-Between Images Through Learned Latent Space Representation Using Variational Autoencoders' (202) 8 IEEE Access 149456, 149458.
[118] Ibid 149463.
[119] Ibid 149457.
[120] Ian Goodfellow et al., 'Generative Adversarial Networks' (2020) 63(11) Communications of the ACM. 139, 140.
[121] Ibid 139.
[122] Ibid139.

generate AIGC consistent with the statistical properties of input data[123] rather than directly reproducing specific training examples. Empirical studies imply that over 99.9 % of GenAI's use would not result in copies in the form of AIGC.[124] Nevertheless, AIGC may still be similar to training data in the following three situations.

First, when the prompt contains text descriptions corresponding to copyrighted images, such as 'draw a Snoopy eating a hamburger', GenAI may generate an image featuring a dog that bears significant resemblance, albeit not identical, to the copyrighted character Snoopy.[125] GenAI encodes the prompt and maps the token 'Snoopy' to the visual concept of Snoopy that it learned during training. Since all the Snoopy images in the training dataset, despite differences in size, color or environment, share a set of features such as a white beagle with black ears and a friendly and relaxed demeanor, GenAI recognizes such features as the statistical patterns representing 'Snoopy'. When presented with a prompt containing a description of 'Snoopy', the GenAI model maps it to these learned features and generates an image probabilistically on the basis of them. Therefore, although the model lacks a human-like understanding of 'Snoopy', it can produce an image with identifiable features of Snoopy because of the patterns it learns while allowing for variations in colors or environmental factors from the training images. Second, users may engage in 'prompt engineering', where they construct detailed descriptions of a copyrighted work to guide the AI toward producing AIGC similar to the work.[126] For example, you can prompt 'draw a white beagle with black ears, a large nose and a red collar, standing and smiling.' You might further add more detail about colors, positions or additional elements to further refine the output and ultimately obtain an image closely resembling the copyrighted character Snoopy. Third, despite specific content in the prompt, similarities can arise owing to overfitting, although this rarely happens. An empirical study by Carlini et al. revealed that among 175 million potential copies generated by GenAI, only 109 could be reasonably considered 'near-copies' of preexisting works.[127] This limited similarity between AIGC and preexisting works has also been corroborated by Henderson et al.[128] Additionally, the study highlighted the necessity of access to costly computational resources, which are typically beyond the reach of average users.[129] As a result, it is rare for AIGC to closely resemble its training data.

As AIGC has potential for public distribution, such AIGC, being substantially similar to prior works, affects copyright owners' economic incentives and satisfies the economic element's requirement. However, some argue that the similarity between AIGC and prior works does not result from GenAI's reproduction of the works[130] 'but out of random by using specific data points that it coincidentally resulted in an output that resembles an input.'[131] Therefore, the question is whether the technological element's requirement is satisfied when AIGC is similar to prior works. We argue that the technological element's requirement is satisfied when a reproduction is made, regardless of the method employed.[132] In the case of human-generated content, as long as the content is similar to a preexisting work, it constitutes an infringement of the reproduction right irrespective of whether the reproducer directly copies the prior work or reverse engineers it through 'decoding-encoding'.[133] In other words, what copyright concerns is the outcome of whether reproductions are made 'regardless of by which means of procedure … they are made'.[134] Just as humans can make reproductions through reverse engineering by 'decoding and encoding', GenAI can also make reproductions through 'encoding, mapping, inference and decoding.' Copyright law has never rejected but rather embraced new means of reproduction. Therefore, in the above three scenarios, GenAI's use of works encompasses the technological element.

Accordingly, in circumstances where the AIGC exhibits substantial similarity to copyrighted works in the training dataset, GenAI's reproduction of works possesses both technological and economic elements, thereby falling within the scope of copyright regulation. As in the preceding three stages, reproductions made during the generation stage may also trigger copyright regulation beyond this stage. For example, someone may publicly distribute AIGC resembling copyrighted works and thereby harm the market for the original works. This potential infringement occurs during the distribution stage rather than the generation stage, with the distributor of the AIGC being the potential infringer, not the creator of the GenAI model. Such infringement is unrelated to the technological essence of GenAI and can potentially arise in the distribution of any object.

In summary, GenAI engages in copyright-regulated reproduction within the generation stage when AIGC is substantially similar to copyrighted works in the training dataset. Table 2 provides a comprehensive overview of the technological and economic elements of GenAI's reproduction at different stages, as well as the circumstances under which copyright-regulated reproduction may occur.

4. Possible criticism

According to Table 2, GenAI generally falls outside the regulation of copyright law. However, it is undeniable that GenAI has significantly benefited from preexisting works. Consequently, our proposition is susceptible to prompting apprehension or inciting critique because it seems to engender an illegal and unfair situation wherein GenAI can profit from preexisting works without paying for them. This section responds to this possible criticism by scrutinizing instances of GenAI's free riding that have been alleged in recent cases and evaluating the legal and moral rationale for prohibiting free riding.

4.1. Free riding

Free riding concerns have been raised by several influential cases involving GenAI. For example, apart from the allegation of copyright infringement, a group of visual artists accused Stability AI, Midjourney and DeviantArt of unlawfully 'appropriating millions of copyrighted

[123] Input data refers to data provided to a system or a model as an input for processing or analysis. In the training phase, the input data is the training data itself. After the model is trained and used by users, the input data also includes new data provided by users or systems, which undergoes the same preprocessing steps as the training data and is used for generating predictions or decisions. Alooba, 'Data Input' https://www.alooba.com/skills/concepts/data-input/ accessed 21 July 2024.

[124] Sag, 'Copyright Safety for Generative AI' (n 15) 326; Murray (n 36) 310.

[125] Sag, 'Copyright Safety for Generative AI' (n 15) 295.

[126] Murray (n 36) 310.

[127] Carlini et al. (n 15) 5258.

[128] Henderson et al (n 13) 8.

[129] We owe a debt to the anonymous reviewer for this observation.

[130] Timothy B. Lee, 'Stable Diffusion Copyright Lawsuits could be a Legal Earthquake for AI' (Ars Technica, 4 March 2023) https://arstechnica.com/tech-policy/2023/04/stable-diffusion-copyright-lawsuits-could-be-a-legal-earthquake-for-ai/ accessed 21 July 2024.

[131] Sandiumenge (n 88) 44.

[132] Pihlajarinne (n 44) 956.

[133] *Altera Corp. v. Clear Logic, Inc.,* 424 F.3d 1079, 1087 (9th Cir. 2005) ('To establish a legitimate reverse engineering defense, Clear Logic must prove by a preponderance of the evidence, the following requirements: Number 1. That Clear Logic reproduced Altera's mask works solely for the purpose of teaching, substantially analyzing, or substantially evaluating non-protectable concepts or techniques embodied in Altera's mask works'). This case explicitly delineates reverse engineering as an act of reproduction, namely, possessing the technological element. When this act of reproduction is undertaken for permitted purposes, it qualifies for an exemption under copyright law. In accordance with our argument, legitimate reverse engineering can be categorized as reproduction with a permitted economic element, which is regulated by exemption. See Section 1.3.

[134] Section 16, Act on Copyright and Related Rights of Germany.

Table 2
GenAI's reproduction involves different stages (the nongray zone represents stage-specific reproduction, and the gray zone denotes reproduction beyond this particular stage).

Stage		technological element	economic element	copyright-regulated reproduction within the stage	copyright-regulated reproduction beyond the stage
data acquisition		yes	no	no	training datasets substituting for original works
Preprocessing		yes	no	no	AIGC resembling training data
training	normal	no	no	no	
	exception	yes (overfitting)			
generation	normal	no	no	no	public distribution of AIGC
	exception	yes	yes	AIGC resembling training data	

images created by artists and using these images as Training images',[135] arguing that such 'misappropriation merely capitalizes on Defendants' theft of Plaintiffs' artistic work and the associated value of Plaintiffs' names.'[136] In *New York Times v. Microsoft, Open AI, etc.,* defendants use millions of articles published by The Times to train their automated chatbots, such as ChatGPT and Bing Chat, which now compete with the news outlet as a source of reliable information.[137] The New York Times complains that defendants' 'misuse and misappropriation of Times content'[138] for training their GenAI models not only 'directly infringed The Times's exclusive rights in its copyrighted works'[139] but also 'constitutes free-riding on The Times's significant efforts and investment of human capital to gather this information'.[140]

Misappropriation is a doctrine that originated from the case of *International News Service v. Associated Press*[141] and is preempted 'to extent it is equivalent to federal copyright law'[142] unless extra elements are found.[143] The key extra element is that 'defendant's use of the information constitutes free riding on the plaintiff's efforts', which 'would so reduce the incentive to produce the product or service that its existence or quality would be substantially threatened'.[144] In fact, owing to the inherent ambiguity of the misappropriation doctrine,[145] courts normally define the elements of misappropriation on the basis of whether the defendant is free riding on the plaintiff's works.[146] Thus, the real issue lies in determining whether free riding is permissible.

4.2. Rationale for prohibiting free riding

A general rejection of free riding lacks a robust foundation. Free riding generally occurs when one benefits from others' property without paying for it.[147] Prohibiting free riding assumes that property right owners have the right to internalize all the benefits generated by their property.[148] However, even tangible property law rejects this assumption.[149] For example, I plant some flowers in my garden, and passersby can also benefit from its fragrance. However, property law does not entitle me to demand a payment from free-riding passing individuals. Property law only allows property owners to recover their costs of investment and obtain reasonable profits, as this is a sufficient incentive for producing and holding property. As the Second Circuit Court of Appeals explicitly stated, only the category of free riding that 'would so reduce[s] the incentive to produce the product or service' can trigger an infringement claim.[150] Although in *L'Oréal v. Bellure* the European Court of Justice considered free riding 'parasitism' and posited that the unfairness stemmed not from the harm done to the plaintiff but from the defendant's attempt to exploit the plaintiff's marketing efforts without offering any financial compensation or putting in any effort themselves,[151] the *L'Oréal* decision is criticized as 'deeply problematic'[152] and 'runs counter to the thrust of earlier ECJ decisions'.[153] What the law intended to prevent is not free riding *per se* but 'the potential lack of any such product or service because of the anticipation of free-riding.'[154] If certain instances of free riding do not undermine the incentive to produce and maintain ownership of a property, such as when consumers derive a surplus value from a property that exceeds its price, known as 'consumer surplus', such instances are not prohibited by law.

Some have objected to free riding from a broader moral standpoint, arguing that free riding 'seems clearly to connote acts that are quintessentially unfair'[155] and perceiving 'something inherently wrong with reaping where another ha[s] sown,'[156] as posited by Locke's labor theory.[157] Since laborers invest time, skill, and resources in their products, they should be entitled to control their use and to benefit from their

[135] *Andersen et al v. Stability AI Ltd. et al* (n 17), Court Docket, Complaint ¶ 144.
[136] Ibid Complaint ¶ 210.
[137] *New York Times v. Microsoft, Open AI, et al*, Case 1:23-cv-11195 Document 1 Filed 12/27/23., Complaint ¶72.
[138] Ibid Complaint ¶197.
[139] Ibid Complaint ¶ 160-164.
[140] Ibid Complaint ¶ 196.
[141] 248 U.S. 215 (1918).
[142] *Fred Wehrenberg Circuit of Theatres v. Moviefone*, 73 F. Supp. 2d 1044 (E.D. Mo. Nov. 1, 1999). *See also Barclays Capital Inc. v. Theflyonthewall.com, Inc.*, 650 F.3d 876, 895 (2d Cir. 2011) ('The broad misappropriation doctrine... is therefore equivalent to the exclusive rights in copyright law.').
[143] *Barclays Capital Inc. v. Theflyonthewall.com, Inc.*, 650 F.3d 876, 893 (2d Cir. 2011).
[144] *National Basketball Association v. Motorola, Inc.*, 105 F.3d 841, 845 (2d Cir. 1997) (holding that 'the surviving "hot-news" INS-like claim is limited to cases where: (i) a plaintiff generates or gathers information at a cost; (ii) the information is time-sensitive; (iii) a defendant's use of the information constitutes free riding on the plaintiff's efforts; (iv) the defendant is in direct competition with a product or service offered by the plaintiffs; and (v) the ability of other parties to free-ride on the efforts of the plaintiff or others would so reduce the incentive to produce the product or service that its existence or quality would be substantially threatened.').
[145] Richard A. Posner, 'Misappropriation: A Dirge' (2003) 40 Houston Law Review 621, 621.
[146] Mark A. Lemley, 'Property, Intellectual Property, and Free Riding' (2005) 83(4) Texas Law Review 1031, 1043.
[147] *International News Service v. Associated Press*, 248 U.S. 215, 239, 39 S.Ct. 68 (1918) (defining free riding as 'taking material that has been acquired by complainant as the result of organization and the expenditure of labor, skill, and money, and which is salable by complainant for money, and ... appropriating it and selling it as [the defendant's] own').
[148] Anne Barron, 'Copyright Infringement, "Free-Riding" and the Lifeworld' in Lionel Bently, Jennifer Davis & Jane C. Ginsburg (eds), *Copyright and Piracy: An Interdisciplinary Critique* (Cambridge University Press 2010) 108.
[149] Lemley (n 146) 1043.
[150] *Barclays Capital Inc. v. Theflyonthewall.com ('Fly I'), Inc.*, 650 F.3d 876, 887 (2d Cir. 2011); *National Basketball Association v. Motorola, Inc.*, 105 F.3d 841, 845 (2d Cir. 1997)).
[151] *L'Oréal SA v. Bellure NV* (C-487/07) 18 June 2009 (ECJ), at 41.
[152] Dev Gangjee & Robert Burrell, 'Because You're Worth It: L'Oréal and the Prohibition on Free Riding' (2010) 73(2) The Modern Law Review 282, 290.
[153] Ibid 293.
[154] *National Basketball Association v. Motorola, Inc.*, 105 F.3d 841, 854 (2d Cir. 1997).
[155] *Barclays Capital Inc. v. Theflyonthewall.com, Inc.*, 650 F.3d 876, 903 (2d Cir. 2011).
[156] Stuart Banner, *American Property: A History of How, Why, and What We Own* (Harvard University Press 2010) 89.
[157] Michael E. Kenneally, 'Misappropriation and the Morality of Free-Riding' (2015)18 Stanford Technology Law Review 289, 302.

commercial exploitation.[158] This argument seems to thwart the desires of free riders to benefit from others' property without sharing the associated costs and ensures that individuals are entitled to the rewards of their own labor.[159] It arguably encapsulates a moral belief that it is justifiable to prevent individuals from profiting from the efforts of others when such individuals have no rightful claim to those rewards.[160]

Nevertheless, regardless of the plausibility of Locke's theory itself,[161] the conclusion that free riding should be prohibited according to Locke remains unpersuasive. Locke's opposition lies distinctly in the act of directly and literally stealing someone else's crops.[162] His critique of appropriating harvests sown by others is more nuanced, primarily focusing on those harvests that can be but have not been appropriated by the laborer:

Whatsoever [the laborer] tilled and reaped, laid up and made use of, before it spoiled, that was his peculiar Right… But if either the Grass of his Inclosure rotted on the Ground, or the Fruit of his planting perished with-out gathering, and laying up, this part of the Earth, notwithstanding his In-closure, was still to be looked on as Waste, and might be the Possession of any other.[163]

Locke explicitly recognized that the rights of laborers are confined to the benefits that they can fully make use of without waste. Should benefits exist that the laborers cannot entirely appropriate, it is reasonable for others to partake in them. Although Locke posits that ' 'tis plain he desired the benefit of another's Pains, which he had no right to',[164] it simply suggests that the laborer has no duty to provide the benefit to another individual.[165] Nevertheless, if the benefit is readily available, the individual has the privilege to take it, and the laborer has no right to prohibit the individual from doing so.[166] Consequently, a general and broad prohibition on free riding is not only legally unfounded, as demonstrated by the aforementioned cases, but also morally unjustified.

Accordingly, the rhetoric of prohibiting free riding in the context of intellectual property is misleading.[167] Intellectual products are inherently noncompetitive. They inherently generate benefits that the right owner cannot capture, which is known as spillover effects. This is precisely the situation where Locke said that reaping what another has sown is permissible. The institutional designs of intellectual property law, such as a limited period of protection and limitations and exceptions of rights, confirm and safeguard these spillover effects. Like tangible property law, intellectual property law prohibits only free riding that undermines the economic incentives of the right owner for creation, such as the public distribution of large-scale reproductions. However, intellectual property laws do not prohibit actions that do not undermine economic incentives, such as reading or reciting works.

4.3. Response

The free riding of GenAI does not constitute the prohibited category of free riding from either a legal or a moral standpoint. Legally speaking, while GenAI may generally copy preexisting works in the technological sense, such copying is an essential part of machine processing and generally does not generate copies of works with potential for public distribution. Consequently, GenAI's free riding, except in the case where AIGC resembles works in the training dataset and which is already regulated by copyright law, would not generate an outcome that can substitute for the original work, nor would it undermine copyright owners' economic incentives; thus, it does not fall under the category of free riding that is prohibited.

Morally speaking, copyrighted works have spillover effects and generate benefits that cannot be fully captured by the copyright owners. GenAI merely appropriates the benefits that would otherwise be wasted. Furthermore, in the majority of cases, GenAI's free riding does not produce output that can substitute for original works to damage the incentive structure of the copyrighted works.

This once again confirms the central theme of this paper—it is the economic element that determines an act of reproduction to be regulated. The neglect of the economic element of reproduction has resulted in our enduring bias against free riding. Prohibiting free riding behaviors that do not affect incentives for right holders would result in an overly broad intellectual property right and hinder re-creative efforts.[168] As Litman notes, the very act of creation in any medium 'is more akin to translation and recombination than it is to creating Aphrodite from the foam of the sea.'[169] The progress of human civilization relies on copyright law's provision of breathing space for free riding, allowing humans to freely learn from previous works. The advent of GenAI marks another milestone in the advancement of human civilization. Just as the development of photography has given rise to new painting genres, the power of GenAI will also propel human creativity to new heights.[170]

5. Conclusion

Upon examining the origin and purpose of the reproduction right, this paper reveals that copyright-regulated reproduction encompasses both technological and economic elements. It not only pertains to the technological generation of a reproduction but also emphasizes its potential to substitute for the original work and impact the economic incentives of the copyright owner. After all, the purpose of copyright law is to provide copyright owners with economic incentives for creation; hence, only actions with potential for impeding these incentives should be subject to copyright regulation. GenAI can generate reproductions of works through data acquisition, preprocessing, training and generation stages. Nevertheless, in the former three stages, the reproductions lack the potential for substituting for original works, thereby lacking the economic element and falling beyond the scope of copyright regulation. Only in the generation stage when the AIGC resembles works in the training dataset does GenAI conduct copyright-regulated reproduction because of the AIGC's potential for public distribution and undermining of copyright owners' economic incentives.

The significance of our reflection on the reproduction right extends beyond the realm of GenAI but also contributes theoretically to elucidating the economic rationale underlying the spectrum of reproduction regulated by presumption, reproduction regulated by exemption, and non-regulated reproduction. Practically, it helps mitigate the chilling effect of copyright on future copy-reliant innovations that heavily rely on and are propelled by copying. In an era where reproduction becomes

[158] Benjamin G. Damstedt, 'Limiting Locke: A Natural Law Justification for the Fair Use Doctrine' (2002) 112 Yale Law Journal 1179, 1181.

[159] Richard J. Arneson, 'The Shape of Lockean Rights: Fairness, Pareto, Moderation' (2005) 22(1) Social Philosophy and Policy 255, 274.

[160] Richard J. Arneson, 'The Principle of Fairness and Free-Rider Problems' (1982) 92(4) Ethics 616, 628.

[161] See, for example, Jeremy Waldron, 'Two Worries about Mixing One's Labour' (1983) 33(130) Philosophical Quarterly 37; James Toomey, 'Property's Boundaries' (2023) 109 Virginia Law Review 131, 175-177.

[162] A. John Simmons, *The Lockean Theory of Rights* (Princeton University Press 1992) 277.

[163] John Locke, *Two Treaties of Government bk. II* (Peter Laslett ed., Cambridge University Press 1999) § 38, at 295.

[164] Ibid § 34, at 291.

[165] Wesley Newcomb Hohfeld, *Fundamental Legal Conceptions As Applied in Judicial Reasoning* (Walter Wheeler Cook ed., Yale University Press 1919) 38–39.

[166] Ibid.

[167] Lemley (n 146) 1032.

[168] Lemley (n 146) 1058.

[169] Jessica Litman, 'The Public Domain' (1990) 39 Emory Law Journal 965, 966.

[170] Lemley & Casey (n 4) 776.

inevitable and ubiquitous, clarifying the boundary of copyright-regulated reproduction safeguards against excessive encroachment of copyright that undermines public interests.

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